Alkane

21. (b) Step 1: Nature of Reaction

- $AlCl_3$ is a Lewis acid catalyst.
- In presence of HBr, it helps in carbocation formation.
- Straight chain alkane can undergo isomerisation + substitution under strong Lewis acidic conditions.

Step 2: Carbocation Formation

Butane ($CH_3 - CH_2 - CH_2 - CH_3$) forms a primary carbocation first, which rearranges to a more stable secondary carbocation.

Primary carbocation:

$CH_3 - CH_2 - CH_2 - CH_2^+$ (unstable)

Rearranges to:

$CH_3 - CH^+ - CH_2 - CH_3$ (secondary, more stable)

Step 3: Attack of Br^-

Br^- attacks the secondary carbocation.

Final Product: 2-Bromobutane

Structure:

$CH_3 - CH(Br) - CH_2 - CH_3$

(Reason: Formation of more stable 2° carbocation before substitution)

22. (b) Ethane is a saturated compound it can not be catalytically hydrogenated.

23 (a) Distillation of petroleum to get different fractions

Explanation:

- Petroleum refining mainly involves fractional distillation.
- Crude oil is heated and separated into different fractions based on boiling points.
- Fractions obtained: LPG, petrol, kerosene, diesel, lubricating oil, bitumen, etc.

Other options:

- (b) → Aromatization is a secondary process, not refining itself.
- (c) → Cracking is also a secondary process.

- (d) → Refining is not simple purification; it mainly means separation into useful fractions.

24 (b) Ethylene dibromide

Explanation:

- Tetraethyl lead (TEL) was added to petrol as an antiknock agent.
- During combustion, lead particles could deposit in the engine.
- To prevent this, ethylene dibromide was added.
- It reacts with lead to form volatile lead bromide (PbBr2), which escapes with exhaust gases.

Other options:

- (a) Iso-octane → Used to measure octane number.
- (c) Tetraethyl lead → Causes lead deposition.
- (d) Mercaptan → Used as odorant in LPG.
- (e) n-Heptane → Used in octane number scale.

25. (a) Branched hydrocarbons are more desirable because they are more volatile.

26. (d) $CH_4 \xrightarrow{Cl_2} CH_3Cl \xrightarrow{Cl_2} CH_2Cl_2 \xrightarrow{Cl_2} CHCl_3 \xrightarrow{Cl_2} CCl_4$.

Since this reaction takes place by free radical mechanism. Hence, there is a possibility of formation of ethane.

$$\underset{\text{Methyl free radicals}}{\dot{C}H_3 + \dot{C}H_3} \rightarrow \underset{\text{Ethane}}{CH_3 - CH_3}$$

27. (c) n-heptane

Concept Used:

Boiling point of alkanes depends on:

1. Molecular mass (↑ mass → ↑ boiling point)
2. Surface area (straight chain > branched chain)
3. Van der Waals forces (stronger in larger molecules)

28. (c) $\underset{\text{Aluminium carbide}}{Al_4C_3} + 6H_2O \rightarrow \underset{\text{Methane}}{3CH_4} + 2Al_2O_3$

29. (d) C_2H_6 — Alkane; no functional group → does not react with PCl_5.

30 (c) Hexane — Alkane; inert and insoluble even in hot conc. H_2SO_4.

31. (a) $C_2H_5O\boxed{H+CH_3}-Mg-Br \rightarrow CH_4 + Mg\begin{matrix} Br \\ OC_2H_5 \end{matrix}$

32. (a) $CH_3I + 2H \xrightarrow{Zn/HCl} CH_4 + HI$

$CH_3I + 2Na + ICH_3 \xrightarrow[\text{Ether}]{\text{Dry}} CH_3 - CH_3 + 2NaI$

33 d) Saturated hydrocarbon — Paraffin wax is a mixture of long-chain alkanes, which are saturated hydrocarbons.

34 d) 1 — Monochlorination of 2-methylbutane forms only one chiral product, giving one pair of enantiomers.

35 a) Aliphatic hydrocarbons — Petroleum mainly contains alkanes and cycloalkanes (non-aromatic hydrocarbons).

36. (a) Solvent for fat, oil, varnish and rubber

37. (d) Synthetic dyes, drugs, perfumes all are made from coal tar.

38. (a) In alkanes, hybridization is sp^3. Hence bond angle is $109^o.5'$.

39. (d) $2CH_3COONa + 2H_2O \xrightarrow{\text{Electrolysis}} CH_3 - CH_3 + 2CO_2 + 2NaOH + H_2$

40. (c) Free-radical substitution — Halogenation of alkanes proceeds via a free-radical chain mechanism (initiation, propagation, termination).